

Niklas Schweiger

M.Sc. Student & Research Assistant · TU Munich

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Master's student at TU Munich and research assistant in Prof. Daniel Cremers' Computer Vision & AI group. Work focuses on inference-time alignment of diffusion and flow models. First-author paper accepted to ECCV 2026.

EDUCATION	M.Sc. Robotics, Cognition, IntelligenceOct 2023 – Present Technical University of Munich (TUM), Germany • Current GPA: 1.6 (German scale) • Focus: Machine learning, generative models, inference-time optimization • Exchange: Chalmers University of Technology, Gothenburg, Sweden (Sep 2024 – Jan 2025)
	B.Sc. Electrical Engineering & Information TechnologyOct 2020 – Sep 2023 Technical University of Munich (TUM), Germany • Final GPA: 2.5 (German scale) • Specialization: Artificial Intelligence and Machine Learning
EXPERIENCE	Research Assistant (HiWi)Mar 2026 – Present Chair for Computer Vision & Artificial Intelligence (CVAI), TU Munich • Developed Trust-Region Noise Search (TRS), a black-box method for aligning diffusion and flow models with arbitrary, non-differentiable reward functions, requiring neither fine-tuning nor gradient access. • Designed and ran evaluations across three domains (text-to-image generation, molecule design, protein design), implementing and benchmarking against multiple baseline methods. • Led the writing of the paper and drove the project to a first-author acceptance at ECCV 2026.
	Internship – AI in Industrial Production2023 Siemens AG Developed a 3D feature matching system using CNN embeddings to automate part retrieval in manufacturing databases, bridging deep learning with industrial applications.
PUBLICATIONS	Schweiger, N., Ram, K., & Cremers, D. (2026). Trust-Region Noise Search for Black-Box Alignment of Diffusion and Flow Models. European Conference on Computer Vision (ECCV), 2026. Also presented at ReALM-GEN Workshop @ ICLR 2026.
AWARDS & ACHIEVEMENTS	2nd Place – TUM.ai MakeathonApr 2023 Built <i>Caire</i>, an app designed to overcome language barriers between nurses and patients using AI to extract medical information from natural speech. Awarded a €2,000 prize.
SKILLS & INTERESTS	Programming Python (Advanced), PyTorch (Advanced), C/C++ (Intermediate), Matlab (Intermediate), LaTeX Languages German (Native), English (C1 – Professional working proficiency) Interests Football, Strength training & running, Guitar (10+ years), AI & Natural sciences